LaTeX packages for R and Advanced knitr

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More ways to combine R and LaTeX

- Additional knitr options for formatting R output: `\Sexpr{}`, `results='asis'`
- `xtable` - formats R tables and data frames as LaTeX tables
- `stargazer` - displaying R models in LaTeX
- `Hmisc` - display generic R objects in LaTeX

Referencing R objects in the LaTeX document

```
x <- 15
y <- rnorm(10, 1)
```

Often, we want to reference numerical results in the text: reporting summary statistics, number of observations, p-values...

we don't want to have to replace those manually.

`\Sexpr{R code}` lets you reference your R data inline.

Referencing R objects in the LaTeX document

`\Sexpr{R code}` lets you reference your R data inline.

For example,

```
$x=\Sexpr{x}$ and $\overline{y}=\Sexpr{mean(y)}$
```

produces the output $x = 15$ and $\overline{y} = 0.5928$ when compiled in a knitr document. As long as the code chunk you are referencing **precedes** the `\Sexpr{}` command, knitr will be able to fill in the blanks for you!

Creating LaTeX code within R

R has lots of packages to produce LaTeX formatted R objects:

- xtable - make nice tables
- stargazer - regression model tables
- Hmisc - make generic R objects into LaTeX-formatted objects
  This one is a bit higher-level
- texreg - model output
- miscFuncs - more latex tables
- reporttools - descriptive statistics

We're only going to talk about the first 3, but there are many other packages out there to do similar things.

Creating LaTeX Tables with xtable

```
library(xtable)
data(iris)
xtable(head(iris))
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.10	3.50	1.40	0.20	setosa
2	4.90	3.00	1.40	0.20	setosa
3	4.70	3.20	1.30	0.20	setosa
4	4.60	3.10	1.50	0.20	setosa
5	5.00	3.60	1.40	0.20	setosa
6	5.40	3.90	1.70	0.40	setosa

Creating LaTeX Tables with xtable

```
?xtable
?print.xtable
print(xtable(head(iris), caption="Iris dataset included with R"),
      include.rownames=FALSE, size="footnotesize")
```

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
5.10	3.50	1.40	0.20	setosa
4.90	3.00	1.40	0.20	setosa
4.70	3.20	1.30	0.20	setosa
4.60	3.10	1.50	0.20	setosa
5.00	3.60	1.40	0.20	setosa
5.40	3.90	1.70	0.40	setosa

Table: Iris dataset included with R

Your Turn

Modify blank.Rnw to do the following:

- Use the `digits` option in `xtable` to display only one decimal for the Sepal measurements
- Change the table caption
- Set the table reference label to "irisdata" and reference it in a sentence outside the code chunk
- Remove the row number from the table display

```
library(xtable)
data(iris)
xtable(head(iris), digits=c(0, 1, 1, 2, 2, 0))
```

stargazer: Latex and R Models

```
x <- seq(0, 10, .5)
y <- x+rnorm(length(x))
data <- data.frame(x=x, y=y)

model <- lm(y~x, data=data)
library(stargazer)
stargazer(model)
```

stargazer: Latex and R Models

	Dependent variable:
	y
x	0.963^{***}
	(0.099)
Constant	0.148
	(0.581)
Observations	21
R^2	0.831
Adjusted R^2	0.823
Residual Std. Error	1.380 (df = 19)
F Statistic	93.720^{***} (df = 1; 19)
Note:	*p<0.1; **p<0.05; ***p<0.01

stargazer: Latex and R Models

```
intercept.model <- lm(y~x, data=data)
nointercept.model <- lm(y~0+x, data=data)

stargazer(intercept.model, nointercept.model, float=F, single.row=TRUE,
          font.size="scriptsize", object.names=T, model.numbers=F)
```

	Dependent variable:	
	y	
	intercept.model	nointercept.model
x	0.963*** (0.099)	0.984*** (0.050)
Constant	0.148 (0.581)	
Observations	21	21
R^2	0.831	0.950
Adjusted R^2	0.823	0.948
Residual Std. Error	1.380 (df = 19)	1.347 (df = 20)
F Statistic	93.720*** (df = 1; 19)	383.100*** (df = 1; 20)
Note:		*p<0.1; **p<0.05; ***p<0.01

Your Turn

Using the iris data:

- Model sepal width by petal width; report the results in a LaTeX table using stargazer
- Add a caption to your table using the `title=` option
- Change the independent variable label to "Petal Width" Hint: `dep.var.labels=""`
- Change the dependent variable label to "Sepal Width" Hint: `covariate.labels=""`
- In the text below the table, report and discuss the correlation between Sepal and Petal Width using `\Sexpr{}`.

```
data(iris)
model <- lm(iris$Sepal.Width~iris$Petal.Width)
?stargazer
```

Remember to turn messages off!

Hmisc: Latex and R Objects

Workhorse function: `latex()`

- Very flexible - handles S3 and S4 classes, lists, data frames, matrices...
- Lots of options for formatting
- Has a preview function
- Other packages are easier to use

Hmisc: Latex and R Objects

```r
library(Hmisc)
my.mean <- function(x){
  sum(x, na.rm=TRUE)/length(x)
}
latex(my.mean, file="")
```

```
my.mean ← function (x)
{
    sum(x, na.rm = TRUE)/length(x)
}
```

Hmisc: Latex and R Objects

Functions can also be done with knitr directly:

```
my.mean
```

function(x) sum(x, na.rm=TRUE)/length(x)

Hmisc: Latex and R Objects

Output specific model information with `latex()`

```
model <- lm(y~x, data=data)
model.tab <- cbind(terms=c("Intercept", "x"),
                   Estimates=round(model$coefficients, 3),
                   SE=round(sqrt(diag(vcov(model))), 3))
rownames(model.tab) <- NULL
latex(model.tab, file="")
```

terms	Estimates	SE
Intercept	0.148	0.581
x	0.963	0.099